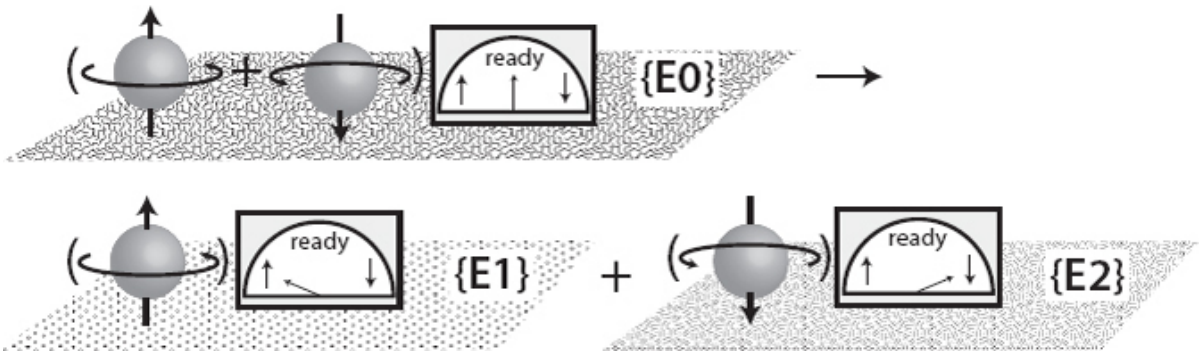


It doesn't really matter what the environment states actually are, so we've portrayed them as different backgrounds labeled {**E0**}, {**E1**}, and {**E2**}. We don't (and generally can't) keep track of exactly what's going on in the environment—it's too complicated. It's not going to just be a single photon that interacts differently with different parts of the apparatus's wave function, it will be a huge number of them. Nobody can be expected to keep track of every photon or particle in a room.

That simple process—macroscopic objects become entangled with the environment, which we cannot keep track of—is decoherence, and it comes with universe-altering consequences. Decoherence causes the wave function to split, or *branch*, into multiple worlds. Any observer branches into multiple copies along with the rest of the universe. After branching, each copy of the original observer finds themselves in a world with some particular measurement outcome. To them, the wave function *seems* to have collapsed. We know better; the collapse is only apparent, due to decoherence splitting the wave function.

We don't know how often branching happens, or even whether that's a sensible question to ask. It depends on whether there are a finite or infinite number of degrees of freedom in the universe, which is currently an unanswered question in fundamental physics. But we do know that there's a lot of branching going on; it happens every time a quantum system in a superposition becomes entangled with the environment. In a typical human body, about 5,000 atoms undergo radioactive decay every second. If every decay branches the wave function in two, that's 2^{5000} new branches every second. It's a lot.